

Regularity results on 3D viscous Tropical Climate Model

joint work with Luca Bisconti (Firenze, IT) and Davide Catania (eCampus, IT)

Diego Berti
(University of Turin, Italy)



UNIVERSITÀ
DI TORINO

AIMS 2024
Evolutionary Equations Systems
19 December 2024, NY University Abu Dhabi

- The **viscous** 3D TCM, for $x \in \mathbb{R}^3$ and $t \geq 0$, is the following system of equations

$$\begin{aligned}\partial_t \mathbf{u} + (\mathbf{u} \cdot \nabla) \mathbf{u} - \Delta \mathbf{u} + \nabla \pi + \nabla \cdot (\mathbf{v} \otimes \mathbf{v}) &= \mathbf{0}, \\ \partial_t \mathbf{v} + (\mathbf{u} \cdot \nabla) \mathbf{v} - \Delta \mathbf{v} + (\mathbf{v} \cdot \nabla) \mathbf{u} + \nabla \theta &= \mathbf{0}, \\ \partial_t \theta + (\mathbf{u} \cdot \nabla) \theta - \Delta \theta + \nabla \cdot \mathbf{v} &= 0, \\ \nabla \cdot \mathbf{u} &= 0,\end{aligned}\tag{TCM}$$

coupled with

$$(\mathbf{u}, \mathbf{v}, \theta)|_{t=0} = (\mathbf{u}_0, \mathbf{v}_0, \theta_0) \quad \text{with} \quad \nabla \cdot \mathbf{u}_0 = 0.$$

Here, the unknowns stand for:

$$\begin{aligned}\mathbf{u} &= (u_1(x, t), u_2(x, t), u_3(x, t)) && \text{barotropic mode of the velocity} \\ \mathbf{v} &= (v_1(x, t), v_2(x, t), v_3(x, t)) && \text{first baroclinic mode of the velocity} \\ \pi &= \pi(x, t) && \text{pressure} \\ \theta &= \theta(x, t) && \text{temperature}\end{aligned}$$

Comparison with other (known) fluid dynamics systems

- If $\mathbf{v} = \mathbf{0}$, it is possible to recover the so-called 3D *Boussinesq Equations* for incompressible viscous flows.

$$\begin{aligned}\partial_t \mathbf{u} + (\mathbf{u} \cdot \nabla) \mathbf{u} - \Delta \mathbf{u} + \nabla \pi &= \theta \mathbf{e}_3, \\ \partial_t \theta + (\mathbf{u} \cdot \nabla) \theta - \Delta \theta &= 0, \\ \nabla \cdot \mathbf{u} &= 0,\end{aligned}\tag{3D Boussinesq Eqs}$$

where $\mathbf{u}$ stands for the **velocity vector field**.

- $\theta = k$, and $\nabla \cdot \mathbf{v} = 0$, then (TCM) closely resembles the 3D *MagnetoHydroDynamic* (MHD) system¹
- If $\theta = k$, and $\mathbf{v} = \mathbf{0}$, one gets the celeb 3D *Navier Stokes* Equations for incompressible viscous flows

¹the difference is in the sign of $\mathbf{v} \cdot \nabla \mathbf{u}$

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$$\begin{aligned}\partial_t \mathbf{u} + (\mathbf{u} \cdot \nabla) \mathbf{u} - \Delta \mathbf{u} + \nabla \pi &= (\mathbf{B} \cdot \nabla) \mathbf{B}, \\ \partial_t \mathbf{B} + (\mathbf{u} \cdot \nabla) \mathbf{B} - \Delta \mathbf{B} &= (\mathbf{B} \cdot \nabla) \mathbf{u}, \\ \nabla \cdot \mathbf{u} = \nabla \cdot \mathbf{B} &= 0,\end{aligned}\tag{3D MHD Eqts}$$

where $\mathbf{v} = \mathbf{B}$ stands for the *magnetic field*.

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- The system (**with no viscosities or dissipations**) originates from the field of **meterology**, specifically focusing on the interaction between large-scale flows and precipitation in the tropical atmosphere. ([Frierson-Majda-Pauluis (2004)]).

COMM. MATH. SCI.

Vol. 2, No. 4, pp. 591–626

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LARGE SCALE DYNAMICS OF PRECIPITATION FRONTS IN THE TROPICAL ATMOSPHERE: A NOVEL RELAXATION LIMIT *

DARGAN M. W. FRIERSON [†], ANDREW J. MAJDA [‡], AND OLIVIER M. PAULUIS [§]

[†]Program in Applied and Computational Mathematics, Princeton University, Fine Hall, Washington Rd., Princeton, NJ 08544 (frierson@math.princeton.edu).

[‡]Department of Mathematics and Center for Atmosphere Ocean Science, Courant Institute of Mathematical Sciences, New York University, 251 Mercer St., New York, NY 10012 (jon-jon@cims.nyu.edu).

[§]Center for Atmosphere Ocean Science, Courant Institute of Mathematical Sciences, New York University, 251 Mercer St., New York, NY 10012 (pauluis@cims.nyu.edu).

Origins of the model & the main question

- The system (**with no viscosities or dissipations**) originates from the field of **meterology**, specifically focusing on the interaction between large-scale flows and precipitation in the tropical atmosphere. ([Frierson-Majda-Pauluis (2004)]).
- The mathematical formulation is derived from a hydrostatic Boussinesq system. In particular, this is achieved **by performing a vertical decomposition of the velocity field** and then applying a **Galerkin truncation** up to the first baroclinic mode.
- Despite originally being in 2D, from the mathematical viewpoint, the system is studied in dimensions $d \geq 2$, with special emphasis on $d = 2$ and $d = 3$, for applications.
- As in other fluid-dynamics systems, a crucial question is understanding **well-posedness**, specifically in terms of **global-in-time existence, uniqueness, and regularity**.

The 2D case

- In the 2D case, [Li-Titi (2016)] showed the global existence of strong solutions, assuming $\mathbf{u}_0, \mathbf{v}_0, \theta_0 \in H^1(\mathbb{R}^2)$.
- Other notable contributions in the 2D case: only dissipation on $\mathbf{v}$, under small initial data [Wan (2016)]; weak fractional viscosity on $\mathbf{u}$ [Z. Ye (2017)]; temperature dependence [X.Ye-Zhu (2019)]: $\nabla \cdot (\mu(\theta) \nabla \mathbf{u})$ in place of $\Delta \mathbf{u}$.

The 3D case

- In the 3D case, while the local well-posedness is known [Ma-Jiang-Wan (2016)], assuming $\mathbf{u}_0, \mathbf{v}_0, \theta_0 \in H^s(\mathbb{R}^3)$, with $s > 3/2$, the **major open question remains**: the global well-posedness of strong solutions.

- To address this issue, various regularization techniques have been explored.

Introduction of damping terms

$$\begin{aligned}\partial_t \mathbf{u} + (\mathbf{u} \cdot \nabla) \mathbf{u} - \Delta \mathbf{u} + |\mathbf{u}|^{\alpha-1} \mathbf{u} + \nabla \pi + \nabla \cdot (\mathbf{v} \otimes \mathbf{v}) &= \mathbf{0}, \\ \partial_t \mathbf{v} + (\mathbf{u} \cdot \nabla) \mathbf{v} - \Delta \mathbf{v} + |\mathbf{v}|^{\beta-1} \mathbf{v} + (\mathbf{v} \cdot \nabla) \mathbf{u} + \nabla \theta &= \mathbf{0}, \\ \partial_t \theta + (\mathbf{u} \cdot \nabla) \theta - \Delta \theta + \nabla \cdot \mathbf{v} &= 0, \\ \nabla \cdot \mathbf{u} &= 0,\end{aligned}\tag{TCMd}$$

where $\alpha, \beta \geq 1$ are the damping coefficients.

If $\theta = k$, and $\mathbf{v} = \mathbf{0}$, $|\mathbf{u}|^{\beta-1} \mathbf{u} \rightarrow$ **Forchheimer drag**, that models the non-linear resistance experienced by a fluid as it flows through a porous medium at *high velocities*.

- For (TCMd), notable contributions include [Yuan-Chen (2021)] and [Yuan-Zhang (2021)], which established the global existence and uniqueness of strong solutions under
 - $\alpha, \beta \geq 4$
 - by incorporating damping in the equation for θ , it suffices for α alone to satisfy $\alpha \geq 4$

large dampings $\rightarrow$ regularizing effect

This suggests that *large damping* coefficients aid in obtaining a priori energy estimates, thus having a regularizing effect.

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Mainly:

- **small initial data** assumptions ([Wang-Zhang-Pan (2020)], [Zhang-Xu (2022)]):
 - **Global well-posedness** with initial data in $(\mathbf{u}_0, \mathbf{v}_0, \theta_0) \in H^2(\mathbb{R}^3)$ such that $\|(\mathbf{u}_0, \mathbf{v}_0, \theta_0)\|_{\dot{H}^{\frac{1}{2}}}$ sufficiently small;
- **increasing the magnitude of viscosity** ([Zhu (2018)], [Yu-Lin-Yin (2022)]):
 - *Fractional diffusion* on $\mathbf{u}$, $(-\Delta)^s \mathbf{u}$, with $s \geq 5/2$: it is proved **global well-posedness of strong solutions**, provided initial data $\mathbf{u}_0, \mathbf{v}_0, \theta_0 \in H^3(\mathbb{R}^3)$;
- **regularity criterion**
 - involving the L^p -norm of $\nabla \mathbf{u}$ [Wang-Zhang-Pan (2020)]:
$$\int_0^T \|\nabla \mathbf{u}\|_{L^p}^q dt < \infty, \text{ with } \frac{2}{q} + \frac{3}{p} \leq 2, 2 < p \leq 3 \Rightarrow \text{the solution is smooth on } [0, T];^2$$
 - in [Wu (2019)], for criterion in terms of Morrey-Campanato norm of spatial derivatives.

²with the flavor of the celeb **Prodi-Serrin condition** for NS Eqts [Prodi (1959)], [Serrin (1962)]

Our main contributions

- In [B.-Bisconti-Catania (2023)], we established an extension criterion for the global well-posedness of strong solutions of (TCMd), in terms of suitable homogeneous Besov norms for (u, v) , in the case $3 \leq \alpha, \beta < 4$.

- Let $3 \leq \alpha, \beta < 4$; $u_0, v_0, \theta_0 \in H^s$, for $s > 3/2$, $\nabla \cdot u_0 = 0$.
- Let (u, v, θ) be a local solution of (TCMd), defined on some interval $[0, T)$, $0 < T < \infty$

Theorem

The local solution can be extended beyond time T , provided that

$$\int_0^T (\|u(t)\|_{\dot{B}_{\infty,\infty}^{\delta}}^{\delta} + \|v(t)\|_{\dot{B}_{\infty,\infty}^{\gamma}}^{\gamma}) dt < \infty$$

where

$$\delta = \delta(\alpha) > 2 \quad \text{and} \quad \gamma = \gamma(\beta) > 2.$$

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Homogeneous Besov quasinorms [book of Bergh-Löfström (1976)]. Using a Littlewood-Paley dyadic decomposition $\{\varphi_j\}_{j \in \mathbb{Z}}$, for $s \in \mathbb{R}$, $1 \leq q \leq \infty$, we can define

$$\|f\|_{\dot{B}_{\infty,q}^s} := \begin{cases} \left(\sum_{j \in \mathbb{Z}} 2^{js} \|\mathcal{F}^{-1} \{\hat{\varphi}_j \hat{f}\}\|_{L^\infty}^q \right)^{1/q}, & \text{if } 1 < q < \infty \\ \sup_{j \in \mathbb{Z}} \{2^{js} \|\mathcal{F}^{-1} \{\hat{\varphi}_j \hat{f}\}\|_{L^\infty}\}, & \text{if } q = \infty, \end{cases}$$

where $\hat{f} = \mathcal{F}(f)$ = Fourier transform of f .

BMO norms. The set $\text{BMO} = \text{BMO}(\mathbb{R}^3)$ is composed by the *bounded mean oscillation functions*, such that is finite

$$\|f\|_{\text{BMO}} := \sup_Q \frac{1}{|Q|} \int_Q |f(x) - f_Q| \, dx,$$

being f_Q the mean of f on Q .

• Fundamental embeddings:

- $\dot{B}_{\infty,2}^0 \subset \text{BMO}$
- $\text{BMO} \subset \dot{B}_{\infty,\infty}^0$

- Why incorporating $\dot{B}_{\infty,q}^0$?
 - **Logarithmic growths in terms of H^s -norms** [Kozono-Ogawa-Taniuchi (2002)]:

$$\|f\|_{\dot{B}_{\infty,2}^0} \lesssim 1 + \|f\|_{\dot{B}_{\infty,\infty}^0} \ln^{1/2}(e + \|f\|_{H^s}), \quad s > \frac{3}{2}. \quad (\text{logarithmic-est})$$

Notation: For $\|f\|_X \lesssim \|f\|_Y$, we mean:

$$\|f\|_X \leq C \cdot \|f\|_Y, \quad C > 0 \text{ independent of } f.$$

- **BMO-Estimate for L^p -Products:**

$$\|fg\|_{L^p} \lesssim \|f\|_{L^p} \|g\|_{\text{BMO}} + \|f\|_{\text{BMO}} \|g\|_{L^p}, \quad p > 1. \quad (\text{BMO-est})$$

Goal: Prove the boundedness of the norm

$$\sup_{\tau \in (0, T)} \|\Lambda^s(\mathbf{u}, \mathbf{v}, \theta)(\tau)\|^2 < \infty,$$

where

$$\|\cdot\| = \|\cdot\|_{L^2}$$

and, for $s > 0$, the operator $\Lambda^s = (-\Delta)^{\frac{s}{2}}$ is defined as

$$\widehat{\Lambda^s f}(\xi) = |\xi|^s \hat{f}(\xi), \quad \xi \in \mathbb{R}^3.$$

Outline of the proof

The proof consists in proving **suitable energy estimates**, using (logarithmic-est), (BMO-est) and a version of the celeb *Kato-Ponce inequality*, as well as the more standard fractional Gagliardo-Nirenberg's and Young's inequalities.

Step 1: L^2 -Estimates.

$$\langle (\text{TCMd}), (\mathbf{u}, \mathbf{v}, \theta) \rangle_{L^2} = 0 \Rightarrow$$
$$\|(\mathbf{u}, \mathbf{v}, \theta)(t)\|^2 + 2 \int_0^t (\|\nabla \mathbf{u}\|^2 + \|\nabla \mathbf{v}\|^2 + \|\nabla \theta\|^2 + \|\mathbf{u}\|_{\alpha+1}^{\alpha+1} + \|\mathbf{v}\|_{\beta+1}^{\beta+1}) = \|(\mathbf{u}_0, \mathbf{v}_0, \theta_0)\|^2.$$

Step 2: H^1 -Estimates.

$\langle (\text{TCMd}), \Delta(\mathbf{u}, \mathbf{v}, \theta) \rangle_{L^2} = 0$, together with **(BMO-est)** and **(logarithmic-est)**, yields:

$$\frac{1}{2} \frac{d}{dt} \|\nabla(\mathbf{u}, \mathbf{v}, \theta)\|^2 + \text{positive terms}$$
$$\lesssim \|\nabla(\mathbf{u}, \mathbf{v}, \theta)\|^2 \left(\mathcal{A} + (\mathcal{A} + \|\mathbf{v}\|_{\dot{B}_{\infty,\infty}^0}^\gamma + \|\mathbf{u}\|_{\dot{B}_{\infty,\infty}^0}^\delta) \ln(e + \|(\mathbf{u}, \mathbf{v})\|_{H^s}) \right),$$

where:

$$\mathcal{A} = \|\mathbf{u}\|_{\alpha+1}^{\alpha+1} + \|\mathbf{v}\|_{\beta+1}^{\beta+1}, \quad \int_0^T \mathcal{A}(\tau) d\tau \leq \|(\mathbf{u}_0, \mathbf{v}_0, \theta_0)\|^2.$$

Intermediate Bound: For every $\varepsilon > 0$, there exists $0 < t_* < t$ such that:

$$\begin{aligned} \|\nabla(\mathbf{u}, \mathbf{v}, \theta)(t)\|^2 + \int_{t_*}^t \text{positive terms} \\ \lesssim \|\nabla(\mathbf{u}, \mathbf{v}, \theta)(t_*)\|^2 e^{\int_{t_*}^t (1 + \|\mathbf{u}\|_{\dot{B}_{\infty, \infty}^0}^\delta + \|\mathbf{v}\|_{\dot{B}_{\infty, \infty}^0}^\gamma) \ln(e + y(\tau)) d\tau} \\ \lesssim C_*(e + y(t))^{C\varepsilon}, \end{aligned}$$

where:

$$y(t) := \sup_{[t_*, t]} \|(\mathbf{u}, \mathbf{v}, \theta)\|_{H^s}, \quad C_* = \|\nabla(\mathbf{u}, \mathbf{v}, \theta)(t_*)\|.$$

Choice of t_* : t_* satisfies:

$$\int_{t_*}^t \|\mathbf{u}(\tau)\|_{\dot{B}_{\infty, \infty}^0}^\delta + \|\mathbf{v}(\tau)\|_{\dot{B}_{\infty, \infty}^0}^\gamma d\tau < \varepsilon.$$

Step 3: H^s -Estimates. Thanks to **Kato-Ponce's inq.**, by applying the operator Λ^s to (TCMd), we obtain:

$$\langle \Lambda^s(\text{TCMd}), \Lambda^s(\mathbf{u}, \mathbf{v}, \theta) \rangle_{L^2} = 0 \Rightarrow$$
$$\frac{d}{dt} \|\Lambda^s(\mathbf{u}, \mathbf{v}, \theta)(t)\|^2 + \|\Lambda^{s+1}(\mathbf{u}, \mathbf{v}, \theta)(t)\|^2 \leq C(t), \quad (1)$$

where:

$$C(t) \approx 1 + \|\nabla \mathbf{u}(t)\|^{\rho_1} + \|\nabla \mathbf{v}(t)\|^{\rho_2} + \|\nabla \theta(t)\|^{\rho_3},$$

and $\rho_i \geq 0$.

Intermediate Result: For any $\varepsilon > 0$, there exists $0 \leq t_* < t < T$ such that:³

$$\int_{t_*}^t C(\tau) d\tau \leq C_*(t - t_*)(e + y(t))^{C_\varepsilon}. \quad (2)$$

Final Step

Combine (1) and (2), and integrate in time to conclude, taking t_* so that C_ε is small enough.

³Reminder:

$$y(t) := \sup_{[t_*, t]} \|(\mathbf{u}, \mathbf{v}, \theta)\|_{H^s}, \quad C_* \approx \|\nabla(\mathbf{u}, \mathbf{v}, \theta)(t_*)\|.$$

A second result in case of small damping: a regularity criterion in terms of BMO space

- Assume $(\mathbf{u}_0, \mathbf{v}_0, \theta_0) \in H^2(\mathbb{R}^3)$, with $\nabla \cdot \mathbf{u}_0 = 0$;
- Suppose that $\boxed{5/3 < \alpha, \beta < 4}$.
- Let $(\mathbf{u}, \mathbf{v}, \theta)(t)$ be the strong solution of (TCMd).

Theorem [B.-Bisconti-Catania (preprint)]

Let $T^* \leq \infty$ be the first blow-up time. Let $X(T^*) \in [0, \infty]$ be defined by

$$X(T^*) := \limsup_{t \rightarrow T^{*-}} \|(\mathbf{u}, \mathbf{v}, \theta)\|_{H^2}.$$

$$X(T^*) < \infty \quad \text{if and only if} \quad \int_0^{T^*} \|\mathbf{u}(t)\|_{\text{BMO}}^8 + \|\mathbf{v}(t)\|_{\text{BMO}}^8 + \|\theta(t)\|_{\text{BMO}}^2 dt < \infty.$$

Case of Fractional Dissipation and Damping

Focus on a modified TCM that incorporates α -fractional diffusion and non-linear β -damping, specifically in the equation for $\mathbf{u}$:

$$\begin{aligned}\partial_t \mathbf{u} + (\mathbf{u} \cdot \nabla) \mathbf{u} + (-\Delta)^\alpha \mathbf{u} + |\mathbf{u}|^{\beta-1} \mathbf{u} + \nabla \pi + \nabla \cdot (\mathbf{v} \otimes \mathbf{v}) &= \mathbf{0}, \\ \partial_t \mathbf{v} + (\mathbf{u} \cdot \nabla) \mathbf{v} - \Delta \mathbf{v} + (\mathbf{v} \cdot \nabla) \mathbf{u} + \nabla \theta &= \mathbf{0}, \\ \partial_t \theta + (\mathbf{u} \cdot \nabla) \theta - \Delta \theta + \nabla \cdot \mathbf{v} &= 0, \\ \nabla \cdot \mathbf{u} &= 0,\end{aligned}\tag{f-TCMd}$$

with $0 < \alpha < 1$.

- We highlight that recently [B.Q. Dong et al. 2019/20], an interpretation has been provided for introducing the α -fractional operator, in place of the classical Laplacian: **lack of density** as altitude increases implies that the **air loses the ability to dissipate heat and diffuse energy**, and nonlocal diffusion seems appropriate,
- The damping balances the (relatively) *weak* regularizing effect of the non-local dissipation term. This serves a primarily technical purpose.

We obtain the following results to shed light on the **interaction between fractional dissipation and damping factor** within the mathematical framework of a model as (f-TCMd).

- Assume $\mathbf{u}_0 \in H^1$, $\mathbf{v}_0, \theta_0 \in H^1(\mathbb{R}^3) \cap L^{2m}(\mathbb{R}^3)$, with $m < 3/(4\alpha - 3)$.

Theorem [B.-Bisconti-Catania (preprint)]

Assume $(\alpha, \beta) \in \mathcal{S}$. The system has a **unique global solution** $(\mathbf{u}, \mathbf{v}, \theta)$ satisfying the following

$$\begin{aligned} & \|\nabla(\mathbf{u}, \mathbf{v}, \theta)\|^2 + \frac{1}{2} \int_0^t (\|(-\Delta)^{\alpha+1} \mathbf{u}\|^2 + \|\Delta \mathbf{v}\|^2 + \|\Delta \theta\|^2) ds \\ & \lesssim \left(\|(\mathbf{u}_0, \mathbf{v}_0, \theta_0)\| + \|\nabla(\mathbf{u}, \mathbf{v}, \theta)\|^2 + \int_0^t (1 + \|\mathbf{v}\|_{2m}^{2m}) ds \right) e^C \int_0^t (1 + \|\nabla(\mathbf{v}, \theta)\|^2) ds. \end{aligned}$$

Definition of the Set $\mathcal{S}$

- The set $\mathcal{S}$ is defined as follows:

$$\mathcal{S} := \left\{ (\alpha, \beta) \in \left(\frac{9}{10}, 1 \right] \times [1, +\infty) : \frac{8\alpha}{6\alpha - 5} \leq \beta + 1 \leq \frac{12}{2\alpha - 1} \right\}.$$

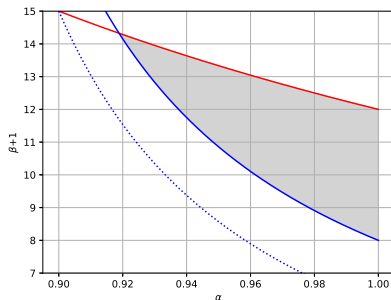


Figure: A zoomed-in view for values $\alpha \in \left[\frac{9}{10}, 1 \right]$ and $\beta + 1 \in [7, 15]$. The set $\mathcal{S}$, defined as the solutions to a system of inequalities (represented by blue/red lines), is shaded in grey.

- We use a direct approach based on energy estimates:

Key Estimates

- L^2 -estimates (uniformly in $t > 0$):

$$\begin{aligned} \mathbf{u} &\in L^2(0, t; \dot{H}^\alpha) \cap L^{\beta+1}(0, t; L^{\beta+1}), \\ \mathbf{v}, \theta &\in L^2(0, t; \dot{H}^1). \end{aligned}$$

- L^{2m} -estimates:

$$\mathbf{v}, \theta \in L^{2m}(0, t; L^{2m}).$$

- H^1 -estimates:

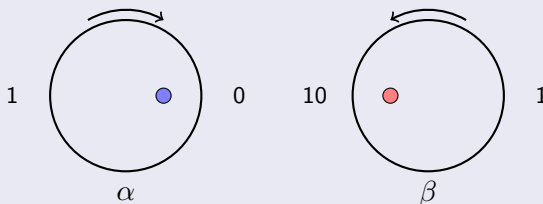
$$\begin{aligned} &\frac{1}{2} \|\nabla(\mathbf{u}, \mathbf{v}, \theta)\|^2 + \frac{1}{2} \|\Lambda^{\alpha+1} \mathbf{u}\|^2 + \frac{1}{2} \|\Delta(\mathbf{v}, \theta)\|^2 + \text{positive terms} \\ &= \int_{\mathbb{R}^3} [(\mathbf{u} \cdot \nabla) \mathbf{u} + \nabla \cdot (\mathbf{v} \otimes \mathbf{v})] \cdot \Delta \mathbf{u} + \int_{\mathbb{R}^3} [(\mathbf{u} \cdot \nabla) \mathbf{v} + (\mathbf{v} \cdot \nabla) \mathbf{u}] \cdot \Delta \mathbf{v} + \int_{\mathbb{R}^3} (\mathbf{u} \cdot \nabla) \theta \cdot \Delta \theta \end{aligned}$$

Main Idea

- **Closing the H^1 -estimates:**

$$\begin{aligned} \frac{1}{2} \|\nabla(\mathbf{u}, \mathbf{v}, \theta)\|^2 + \frac{1}{2} \|\Lambda^{\alpha+1} \mathbf{u}\|^2 + \frac{1}{2} \|\Delta(\mathbf{v}, \theta)\|^2 \\ \lesssim \|\nabla \mathbf{u}\|^* \|\mathbf{u}\|_{\beta+1}^2 + \|\nabla(\mathbf{v}, \theta)\|^* + \|\mathbf{u}\|_{\beta+1}^{\beta+1} + \|\mathbf{v}\|_{2m}^{2m}. \end{aligned}$$

treating α and β as loosely interconnected knobs; any loss in one is compensated by a gain in the other.



Interpretation of $\mathcal{S}$

- The set $\mathcal{S}$ is the intersection of several inequalities directly involving α and β .

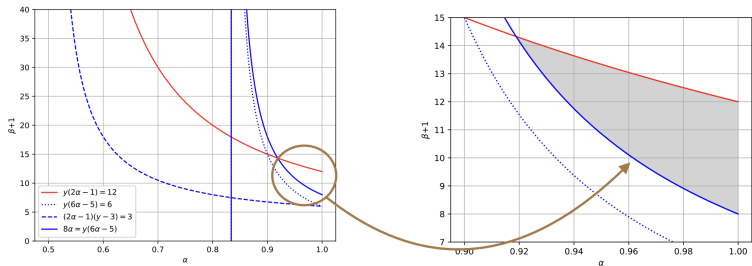


Figure: On the left: the lines represent sufficient conditions in terms of α and $\beta + 1$, ensuring the validity of certain interpolations.

Example Condition

- The area **below the red line** satisfies the inequality:

$$(\beta + 1)(2\alpha - 1) \leq 12 \implies \left| \int_{\mathbb{R}^3} (\mathbf{u} \cdot \nabla) \mathbf{u} \cdot \Delta \mathbf{u} \, dx \right| \leq C \|\nabla \mathbf{u}\|^\delta \|\mathbf{u}\|_{\beta+1}^2 + \varepsilon \|\Lambda^{1+\alpha} \mathbf{u}\|^2.$$

We add **linear** dampings on $\mathbf{v}$ and θ :

$$\begin{aligned}\partial_t \mathbf{u} + (\mathbf{u} \cdot \nabla) \mathbf{u} + (-\Delta)^\alpha \mathbf{u} + |\mathbf{u}|^{\beta-1} \mathbf{u} + \nabla \pi + \nabla \cdot (\mathbf{v} \otimes \mathbf{v}) &= \mathbf{0}, \\ \partial_t \mathbf{v} + (\mathbf{u} \cdot \nabla) \mathbf{v} - \Delta \mathbf{v} + (\mathbf{v} \cdot \nabla) \mathbf{u} + \nabla \theta + \sigma_1 \mathbf{v} &= \mathbf{0}, \\ \partial_t \theta + (\mathbf{u} \cdot \nabla) \theta - \Delta \theta + \nabla \cdot \mathbf{v} + \sigma_2 \theta &= 0, \\ \nabla \cdot \mathbf{u} &= 0,\end{aligned}\tag{f-TCMsd}$$

where $\sigma_1, \sigma_2 > 0$.

Theorem [B.-Bisconti-Catania (preprint)] Assume that $(\alpha, \beta) \in \mathcal{S}$; let $(\mathbf{u}, \mathbf{v}, \theta)(t)$ be the global solution of (f-TCMsd).

Assume also that $\sigma_1, \sigma_2 > \tilde{C} > 0$, where $\tilde{C}$.

We have

$$\lim_{t \rightarrow \infty} (\|\mathbf{u}(t)\| + \|\mathbf{v}(t)\| + \|\theta(t)\|) = 0.$$

- We extend and generalize previous works (see e.g. [Zeng-Yang (2023)]) in the context of a system including fractional dissipation and damping.
- Our method applies *Fourier splitting* and generalized energy inequalities, requiring non-trivial calculations due to the fractional diffusion on $\mathbf{u}$ and the fact that $\mathbf{v}$ is not divergence-free.
- This method, which dates back to [Schonbek (1985)], is based on studying the norm of the Fourier transform of $\mathbf{u}$:⁴

$$\hat{\mathbf{u}}(\xi) = \mathcal{F}\mathbf{u}(\xi) = (2\pi)^{-3/2} \int_{\mathbb{R}^3} e^{-i\xi x} \mathbf{u}(x) dx \quad \text{for } \xi \in \mathbb{R}^3$$

and split the analysis between *small* frequencies ξ and *large* frequencies ξ .

⁴since, from Plancharel's formula $\|\hat{\mathbf{u}}\| = \|\mathbf{u}\|$

Thanks for the attention